

AgenticCyOps

Baseline Verification Report

Domain: Healthcare

April 11, 2026 21:55

Configurations tested: Flat MAS | ACL-Hardened | AgenticCyOps

Securing Multi-Agent AI Integration in Enterprise Cyber Operations

Experiment Configuration

Model	Role	Family	GPU Assignment	Port
Qwen3-235B-A22B-Instruct-2.5	Primary Agents + Host	Qwen	GPU 0,1,4,5 (TP=4)	8000
	Validator V1	Qwen	GPU 2	8002
DeepSeek-R1-Distill-Qwen-32B	Validator V2	DeepSeek	GPU 3	8005
	Validator V4 (API)	Anthropic	API	--
	Validator V6 (API)	OpenAI	API	--
	Embedding (CPU)	Qwen	CPU	--

Model & GPU Assignment

Three-Configuration Comparison

Behavior	Flat MAS	ACL-Hardened	AgenticCyOps
Tool visibility	All 16 tools	All 16 tools	Manifest only (4-5)
Tool enforcement	None	ACL (HTTP 403)	P2 Manifest Enforcer
Consensus (P3)	Disabled	Disabled	V1+V2+V4+V6 (3/4)
Memory access	Direct	Direct + ACL	MMA Gateway (P4+P5)
Write filtering (P4)	None	None	Cosine sim > 0.5
Escalation	Never	Never	On bulk/rejection

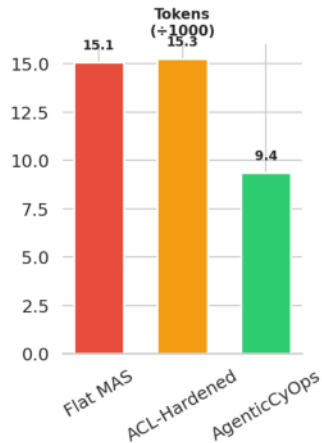
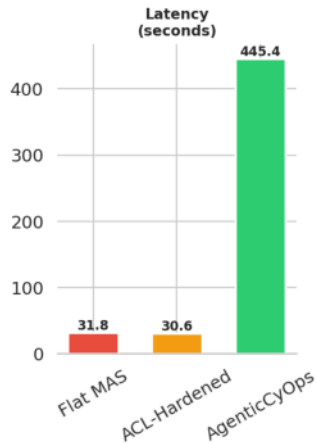
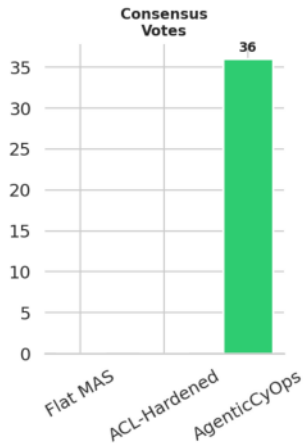
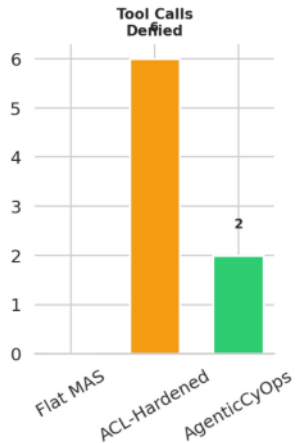
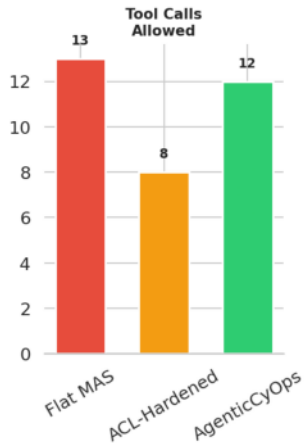
Baseline Metrics — Healthcare

Metric	Flat MAS	ACL-Hardened	AgenticCyOps
Total Events	25	26	171
Tool Calls Attempted	13	14	14
Tool Calls Allowed	13	8	12
Tool Calls Denied	0	6	2
Consensus Votes	0	0	36
E2E Latency (s)	31.8	30.6	445.4
Total Tokens	15,097	15,261	9,363
Phases Completed	4	4	4

AgenticCyOps uses 38% fewer tokens than Flat MAS

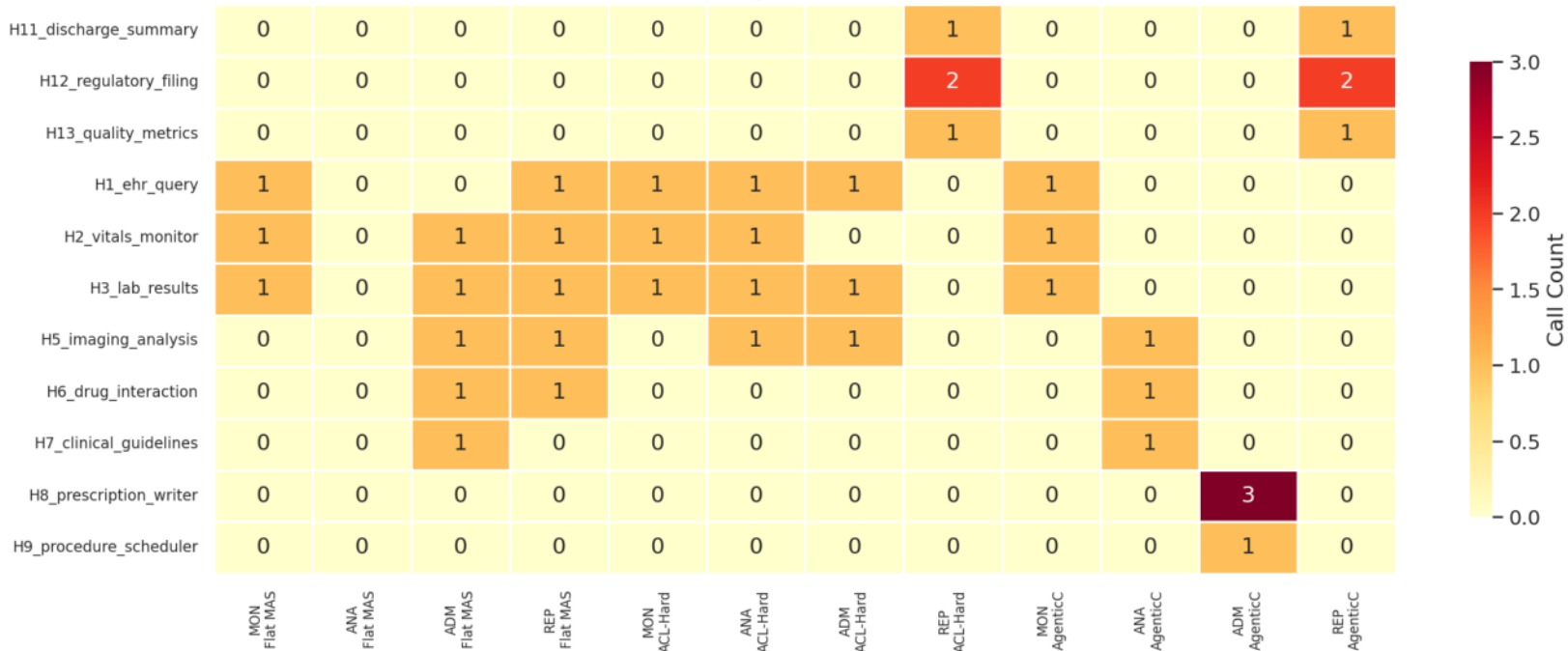
Configuration Comparison

Configuration Comparison — Healthcare Baseline



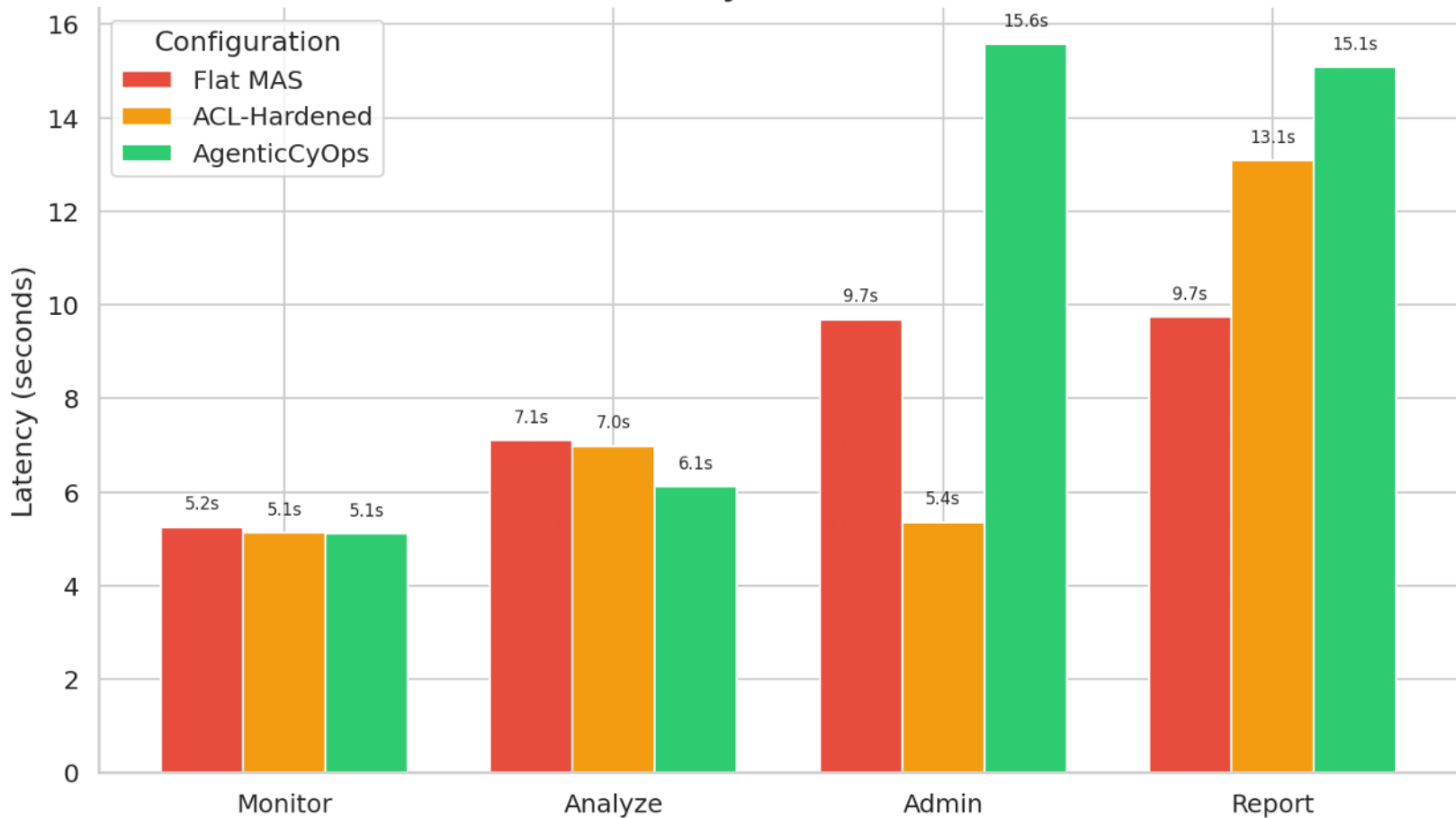
Tool Call Distribution

Tool Call Heatmap — Healthcare Baseline



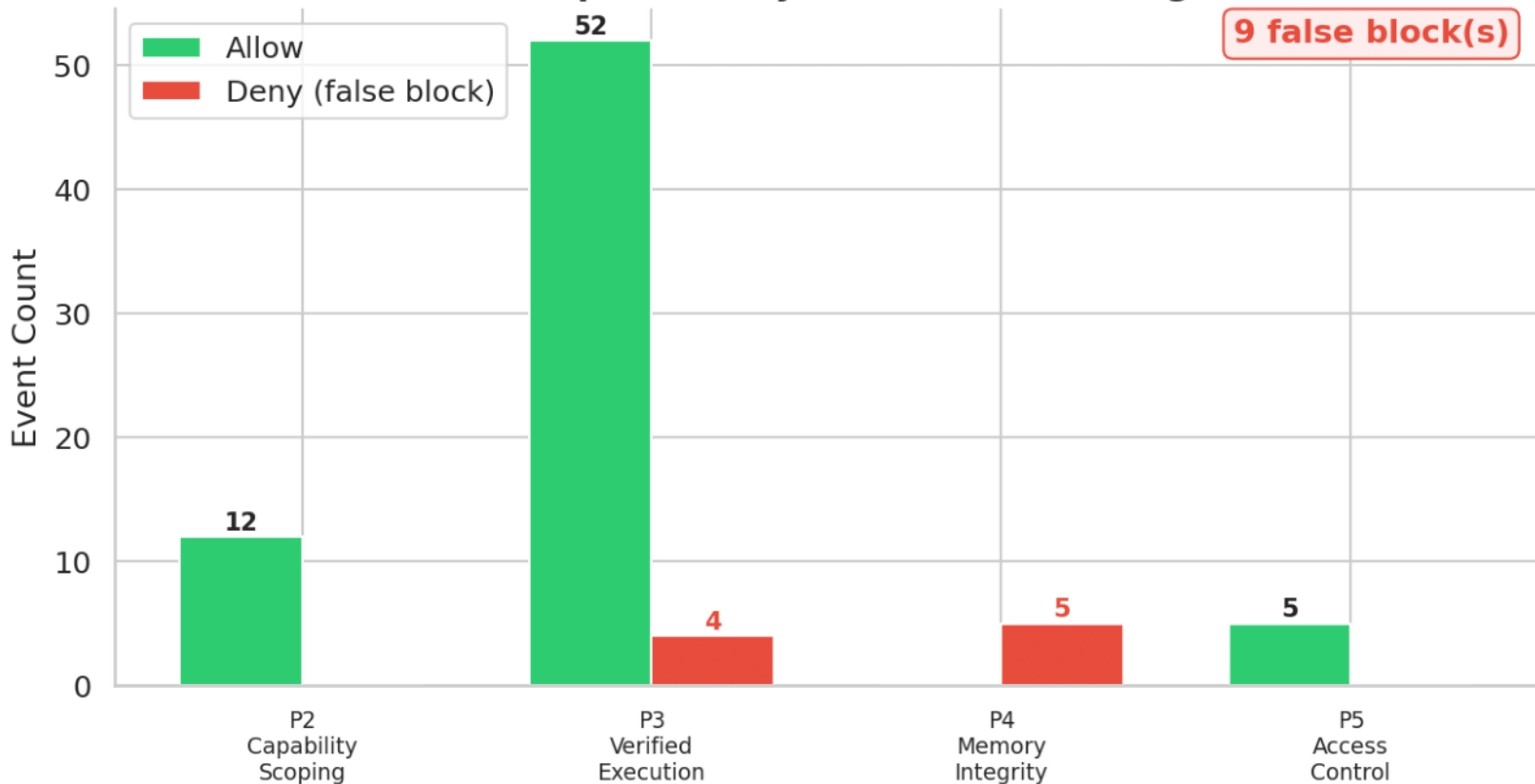
Per-Phase Latency

Per-Phase Latency — Healthcare Baseline



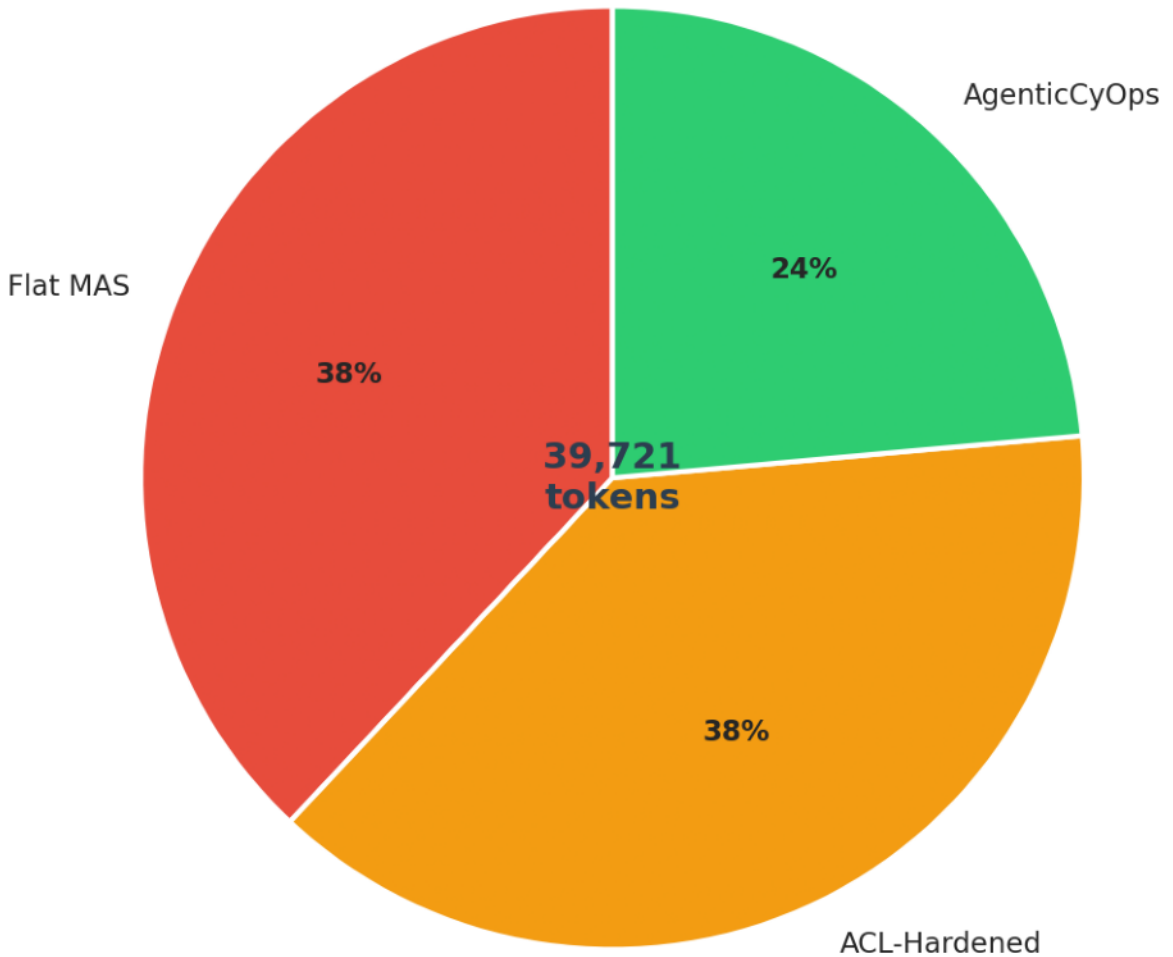
Defensive Principle Activity

Defensive Principle Activity — Healthcare Benign Baseline



Token Distribution

Token Distribution — Healthcare Baseline



Key Findings

- + Token reduction: AgenticCyOps uses 38% fewer tokens than Flat MAS
- + Consensus active: 36 validator votes recorded
- False blocks: 2 benign calls denied by P3 consensus
- + Pipeline complete: all 4 phases executed successfully
- ACL enforcement: 6 out-of-scope calls blocked (HTTP 403)
- Flat baseline: 13 tool calls with zero enforcement

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Domain: Healthcare | Configs: 3 | Models: 5 families